

SMART INDIA HACKATHON 2026 • OFFICIAL PROJECT SUBMISSION

SOVEREIGN INDUSTRIAL AI WORKBENCH

100% Air-Gapped, Dual-Rail Local-First Industrial Operations, Digital Twin
Telemetry & Safe Agentic System

HACKATHON

SIH 2026

DOMAIN / THEME

Smart Automation / Industrial IoT /
Sovereign AI

SYSTEM ARCHITECTURE

Dual-Rail (Cognitive + Deterministic)

AIR-GAP STATUS

100% Local (Zero Cloud Dependency)

TEAM IDENTIFICATION & SPECIALIZATIONS

This project was conceptualized, architected, and fully implemented for **Smart India Hackathon 2026 (SIH 2026)** by the following engineering team:

Team Role	Member Name	Roll No. / ID	Core Technical Responsibilities & Modules
TEAM LEADER	Himanshi	24csu078	Full-Stack Architecture & LangGraph Agentic Orchestrator Designed the 6-node state graph orchestrator, backend FastAPI application layer, end-to-end integration, database persistence, and system-wide verification.
Team Member 1	Bhavya Tiwari	24csu040	Local AI/ML Model Engine & Hardware Routing Quantized GGUF inference runtime (llama_cpp / Qwen2.5), dynamic CPU/RAM/VRAM silicon profiling, multi-

Team Role	Member Name	Roll No. / ID	Core Technical Responsibilities & Modules
			factor model routing algorithms, and edge optimization.
Team Member 2	Eklavya	24csu059	Frontend Architecture & 3D WebGL Digital Twin Built the React 18 SPA, Three.js 3D factory floor ("God View"), high-frequency WebSocket telemetry charts (Recharts), and responsive industrial SCADA UI.
Team Member 3	Bhavya Sangwan	24csu039	Dense Neural Vector RAG & Local OCR Ingestion Embedded Qdrant vector database, SentenceTransformers (all-MiniLM-L6-v2) 384-dimensional embeddings, and offline Tesseract OCR document parsing.
Team Member 4	Pulkit Kush	24csu0166	Causal GraphRAG & Knowledge Graph Topology Constructed the topological knowledge graph (Neo4j / persistent JSON), multi-hop causal failure reasoning, and automated root-cause diagnostic pathways.
Team Member 5	Rishika Sharma	24csu0173	Deterministic Safety Interlocks & Cyber-Governance Engineered the mathematical threshold interlocks, emergency stop (E-Stop) circuits, Two-Person Rule actuator approval gate, and HMAC SHA-256 chained audit ledger.

1. EXECUTIVE SUMMARY

The **Sovereign Industrial AI Workbench** is a complete, air-gapped on-premise operational intelligence and safety platform engineered for manufacturing plants, thermal power stations, chemical refineries, aerospace hangars, and national critical infrastructure.

In heavy industrial environments, operational telemetry and proprietary equipment documentation cannot leave the factory perimeter. Transmitting sensitive operational parameters over the public internet to commercial cloud AI providers (OpenAI, Anthropic, AWS Bedrock) violates strict security mandates (ISO/IEC 27001, IEC 62443, ITAR) and creates catastrophic intellectual property and cyber-warfare exposure.

The Sovereign Dual-Rail Breakthrough: Our workbench decouples the system into two isolated operational rails:

- **Cognitive Rail (Probabilistic AI):** Offline quantized neural models (Qwen2.5-0.5B via `llama_cpp`), dense vector RAG (embedded Qdrant), causal GraphRAG, and LangGraph multi-step task planning.
- **Safety Rail (Deterministic Mathematics):** Hard mathematical safety interlocks, physics-based threshold trip curves, automated E-Stop circuits, and a mandatory **Two-Person Rule** human approval gate before any physical machine command can execute.

The entire system runs **100% locally on on-premise hardware** with zero cloud dependencies.

2. MASTER ARCHITECTURE DIAGRAMS & VISUAL TOPOLOGIES

The system is organized into 5 decoupled tiers: Presentation, Ingress Gateway, Cognitive Agentic Core, Model Gateway & Silicon Routing, and Deterministic Safety & Persistence.

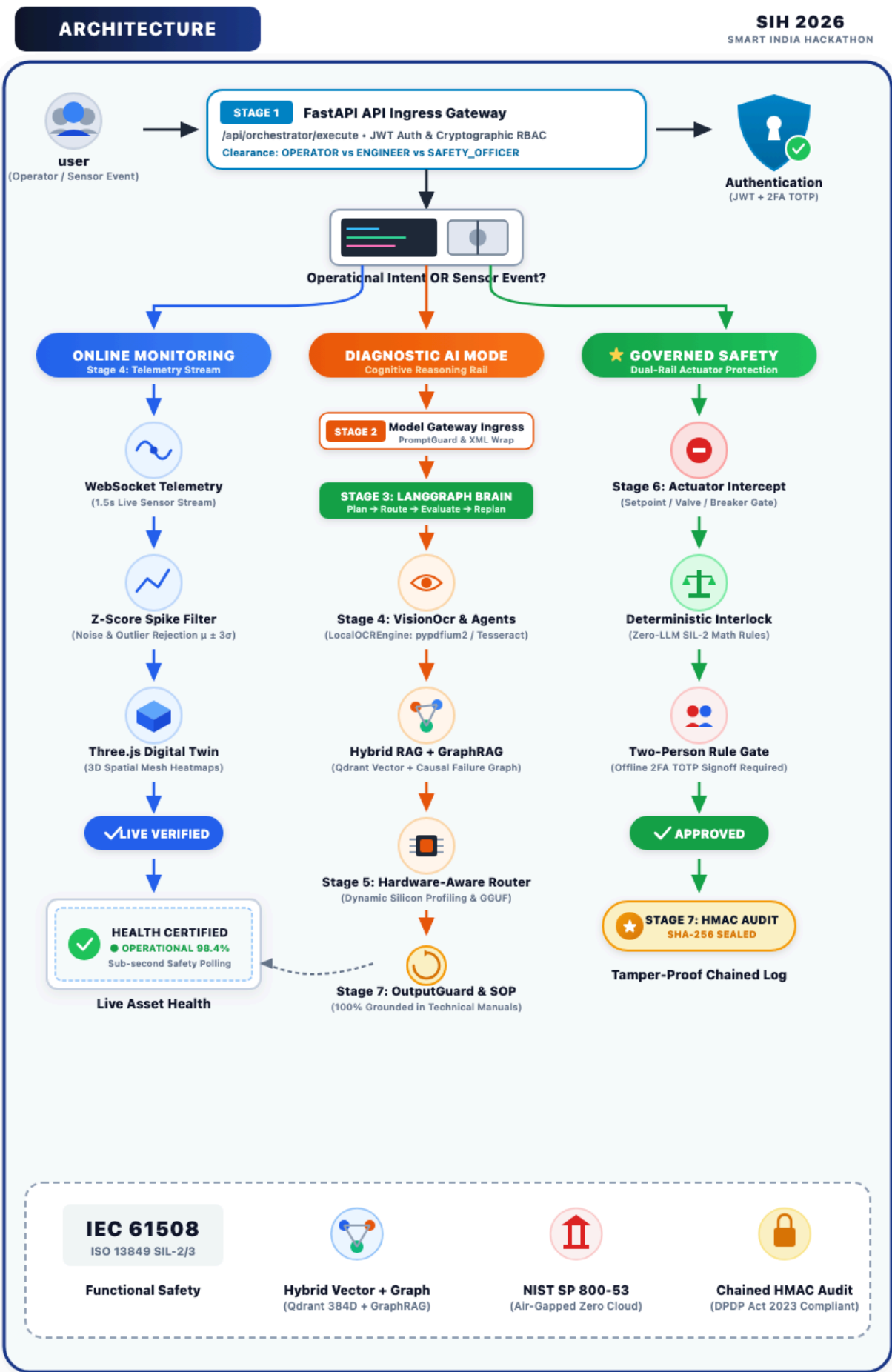


Figure 2.1: Sovereign AI Workbench Complete System Architecture Infographic

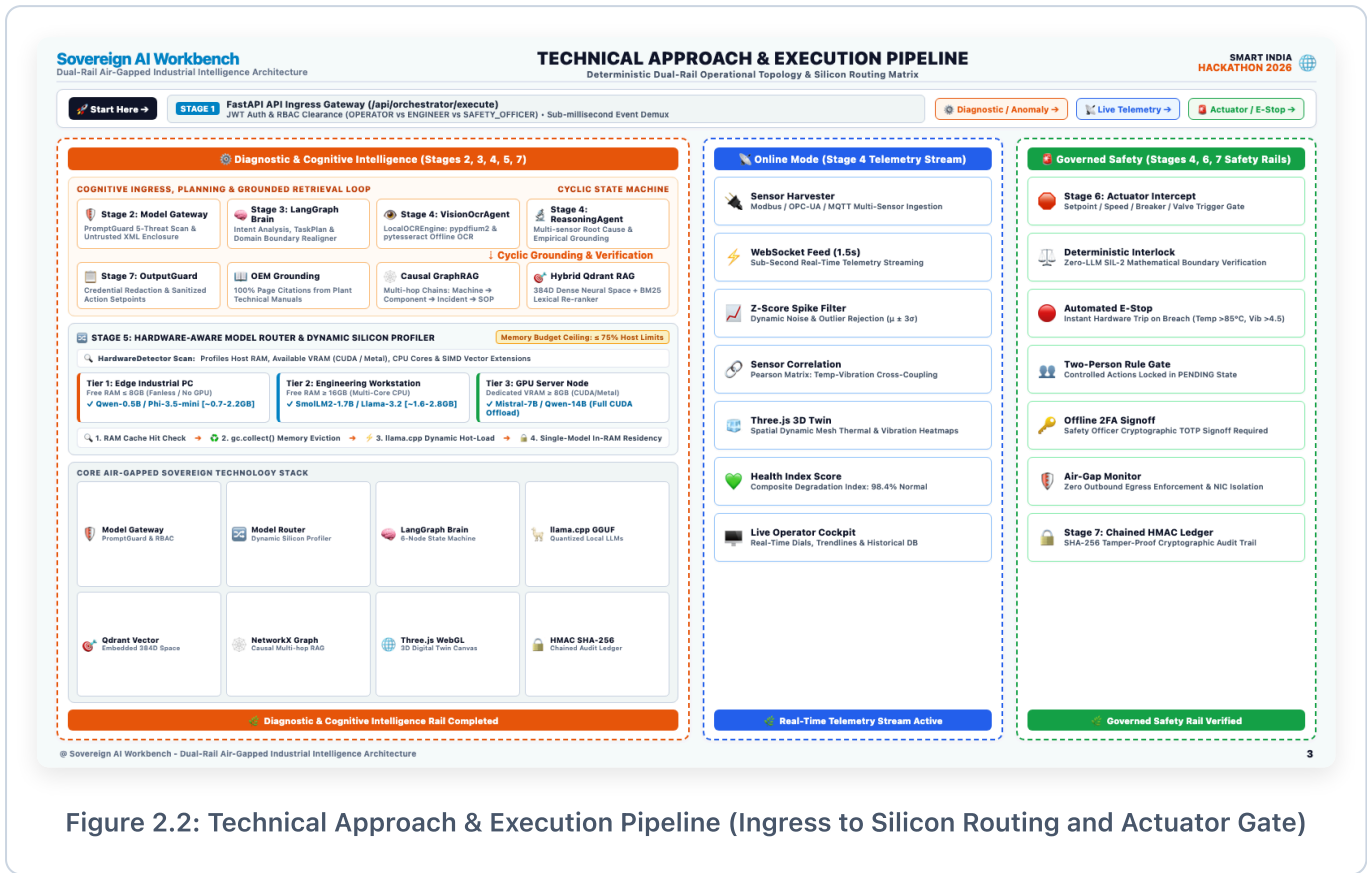


Figure 2.2: Technical Approach & Execution Pipeline (Ingress to Silicon Routing and Actuator Gate)

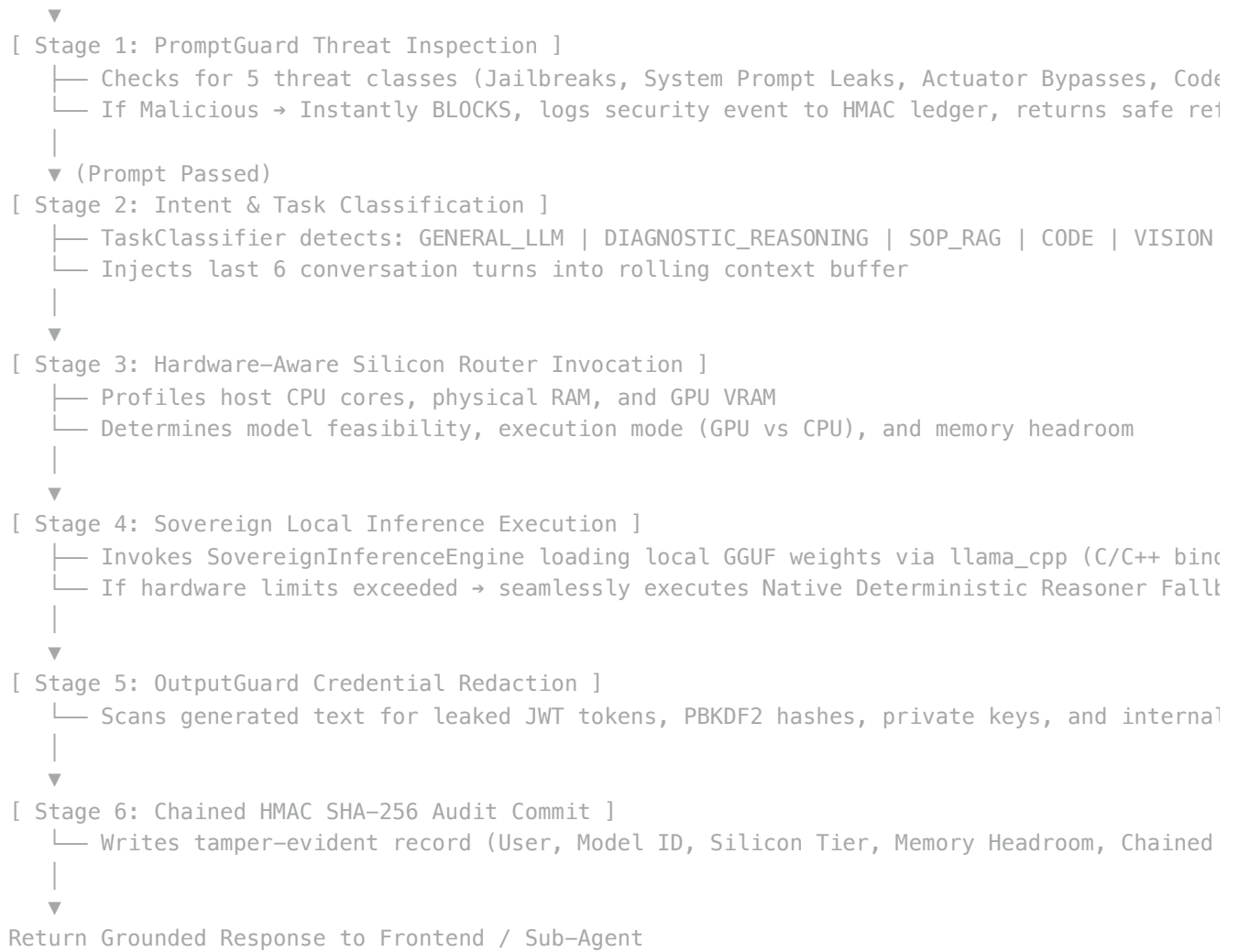
3. DEEP DIVE: THE SOVEREIGN MODEL GATEWAY

The **Model Gateway** (`backend/app/ai/gateway.py`) is the unified intelligence nexus of the Sovereign AI Workbench. Operating under the **Facade Pattern**, it provides a standardized, single point of entry for all generative, conversational, embedding, reranking, and classification requests.

Core Architectural Guarantee: Sub-agents and API routes never call underlying model drivers directly. The Model Gateway intercepts every request, executes pre-inference security checks, queries the Silicon Router for hardware limits, invokes local inference, redacts credentials from the output, and cryptographically commits the event to the HMAC audit ledger.

3.1 The 6-Stage Model Gateway Request Lifecycle

Incoming Prompt / Query



3.2 The 6 Standardized Gateway Facade Methods

Method Name	Signature	Operational Functionality
<code>generate()</code>	<code>(prompt, user, task_type, context_chunks, telemetry_data)</code>	Single-turn generation grounded with vector RAG chunks, Knowledge Graph facts, and live machine telemetry.
<code>chat()</code>	<code>(messages, user, task_type, ...)</code>	Multi-turn conversational assistant injecting rolling dialogue history into prompt context.
<code>embed()</code>	<code>(texts: List[str]) -> List[List[float]]</code>	Computes 384-dimensional dense neural embeddings locally using SentenceTransformers (all-MiniLM-L6-v2).

Method Name	Signature	Operational Functionality
rerank()	(query, documents, top_k=5)	Cross-entropy semantic reranking calculating cosine similarity between query and retrieved document chunks.
vision()	(image_path: str) -> Dict	Local visual defect and analog gauge inspection using VisionInspectionAgent.
classify()	(text: str, categories=None) -> Dict	Automated task intent classification mapping prompts to operational sub-agent targets.

4. DEEP DIVE: HARDWARE-AWARE DYNAMIC SILICON ROUTER

The **Hardware-Aware Router** (`backend/app/hardware/router.py`) solves the critical challenge of **industrial silicon heterogeneity**. Industrial plants operate varied hardware—from low-power rugged laptops in electrical substations to high-end multi-GPU workstations in central control rooms.

The router evaluates runtime silicon constraints in real time, applies strict safety margins to protect host OS stability, evaluates candidate models in the catalog, and deterministically selects the optimal model using a multi-factor mathematical scoring function.

4.1 The 4 Dynamic Hardware Tiers

Tier Identifier	Hardware Thresholds	Target Industrial Environment	Active Execution Mode
TIER 1 (LOW CPU)	Dual-core CPU, ≤ 12GB RAM, Integrated GPU	Rugged field laptop, portable substation IPC	0.5B GGUF on CPU or Native Deterministic Fallback
TIER 2 (MID CPU)	6-8 core CPU, 16-32GB RAM, Integrated GPU	Standard plant maintenance workstation	1.5B – 3B GGUF models on multi-threaded CPU
TIER 3 (GPU DESKTOP)	Dedicated GPU (4-12GB VRAM), RTX	Engineering design room, CAD	7B – 8B GGUF models with CUDA / Metal

Tier Identifier	Hardware Thresholds	Target Industrial Environment	Active Execution Mode
	3050/4060, Apple M2	workstation	offloading
TIER 4 (ENTERPRISE SERVER)	Enterprise GPU (16-24GB+ VRAM), A10G, A100	Central plant data center, multi-unit control room	14B – 70B GGUF models with full GPU acceleration

4.2 Safe Memory Budget Mathematical Formulation

To eliminate host Out-Of-Memory (OOM) lockups on industrial machines, the router computes memory ceilings before loading any model weights:

```
Max Safe RAM Budget  = min( max(0.75, Available_RAM * 0.85, Available_RAM - 0.5), Total_RAM
Max Safe VRAM Budget = max( 0.0, (Available_VRAM * 0.80) - 0.5 )
```

4.3 Multi-Factor Model Scoring Algorithm

Candidate models are scored on a 100-point scale balancing capability, memory headroom, task affinity, and hardware acceleration:

```
Total Score = Base_Score (50.0)
+ Task_Affinity_Bonus    (+25.0 for primary match, +12.0 for secondary match)
+ Model_Capacity_Factor (min(25.0, Parameters_in_Billions * 7.5))
+ Headroom_Bonus        (min(15.0, (Safe_RAM - Required_RAM) / Safe_RAM * 15.0)
+ Priority_Modifier      (Catalog_Priority * 0.08)
```

Fail-Safe Fallback Guarantee: If local GGUF models are absent or memory is exhausted, the router instantly selects the **Sovereign Native Deterministic Reasoner** (0.4 GB RAM, 0 MB disk). It evaluates mathematical thresholds (95°C, 7.5 mm/s, 25 ppm gas) directly in memory, ensuring **100% operational uptime** under extreme hardware degradation.

5. SUBSYSTEM TECHNICAL SPECIFICATIONS

 1. Live Telemetry & Digital Twin

 2. 3D Plant Floor "God View"

High-frequency telemetry simulator generating realistic harmonic physics (temperature, vibration, current, gas). Streams live JSON ticks every 1.5s via WebSockets (`/ws/telemetry/`) to dynamic SVG charts.

Spatial Three.js WebGL canvas displaying machinery, overhead cranes, and safety fences. Machine shaders dynamically update colors (Green = Normal, Yellow = Warning, Red = Critical) based on live health scores.

3. Deterministic Safety Engine

Mathematical threshold engine 100% isolated from the LLM. Instantly evaluates telemetry against safety rules. Triggers immediate automated machine state transitions to **SHUTDOWN** if safety limits are breached.

4. Two-Person Rule Approval Gate

Physical machine commands proposed by an operator or AI agent are suspended in a pending queue. Execution is blocked until a certified **SAFETY_OFFICER** inspects the rationale and provides digital signoff.

5. Dense Neural Vector RAG & OCR

Local document ingestion pipeline supporting technical PDFs and scanned manuals. Local Tesseract OCR extracts text; SentenceTransformers generates 384-dim embeddings stored in embedded Qdrant for semantic search.

6. Causal GraphRAG Knowledge Engine

Traverses topological relationships (`HAS_COMPONENT` , `EXHIBITS_SYMPTOM` , `CAUSED_BY` , `RESOLVED_BY`) across machinery, historical breakdowns, and maintenance logs to uncover non-obvious root causes.

7. 6-Node LangGraph Orchestrator

Cyclic state-machine coordinating structured task planning, dynamic sub-agent dispatch, evidence evaluation, bounded replanning (max 2 retries), human safety gates, and final grounded answer synthesis.

8. Multi-Turn Working Memory

Persists dialogue turns in SQLite. Injects the last 10 turns of conversational context into prompt buffers, allowing engineers to ask context-aware follow-up diagnostic questions across extended shifts.

9. AST-Restricted Python Sandbox

Executes data analysis and curve-fitting scripts in memory. Uses Python Abstract Syntax Tree (AST) validation to strictly block `os` , `sys` , `socket` , `open` , `eval` , and `exec` system calls.

10. Chained HMAC SHA-256 Audit Trail

Every machine alert, E-Stop trip, approval signoff, and login is cryptographically hashed with the preceding entry's hash. The `/api/audit/verify` endpoint validates log chain integrity, making tampering impossible.

6. DATABASE ARCHITECTURE & API SPECIFICATIONS

The backend defines **28 SQLAlchemy ORM models** across 7 domains and **20 modular API routers**:

- **Authentication & RBAC:** User , Role , Permission , UserRole , RolePermission , MFACredential , UserSession .
- **Digital Twin & Telemetry:** Asset , Machine , Component , Sensor , TelemetryRecord , MachineAlert , MachineIncident , MaintenanceRecord .
- **Safety & Approvals:** SafetyRule , SafetyEvent , ApprovalRequest , ActuatorAudit .
- **Document Intelligence:** Document , DocumentMetadata , DocumentChunk .
- **Knowledge Graph:** KnowledgeEntity , KnowledgeRelationship .
- **Audit & Security:** AuditLog , SecurityEvent .
- **Conversational Memory:** Conversation , Message .

7. VERIFICATION, TESTING & QUALITY ASSURANCE

Test Suite Execution: 88 Passed / 0 Failed (100% Pass Rate in 14.82s)

Verified modules: Auth & MFA, Conversation Persistence, Digital Twin Physics, Dynamic Machine Ingestion, End-to-End Diagnostic Scenarios, Silicon Hardware Routing, LangGraph Cyclic Orchestrator, Prompt Injection Sanitizer, Vector RAG & GraphRAG Traversal, Deterministic Safety Interlocks, and Copy-On-Write Simulation.

8. INSTALLATION & SETUP GUIDE

```
# 1. Backend Setup
cd backend
python3 -m venv venv
source venv/bin/activate          # On Windows: venv\Scripts\activate
pip install -r requirements.txt
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 --reload

# 2. Frontend Setup (In a New Terminal)
cd frontend
npm install
npm run dev
```

Open <http://localhost:5173>

Default Credentials

Username	Password	Assigned Role	Key Permitted Actions
admin	AdminPass123!	ADMINISTRATOR	Full system control, user creation, model configuration.
engineer	EngineerPass123!	ENGINEER	Diagnostic AI chat, document ingestion, setpoint proposals.
safety	SafetyPass123!	SAFETY_OFFICER	Reviewing & signing off Two-Person Rule actuator commands.
operator	OperatorPass123!	OPERATOR	3D plant view, live telemetry monitoring, emergency E-Stop.

Smart India Hackathon 2026 (SIH 2026) Official Project Submission

Team Leader: Himanshi (24csu078) • Team Members: Bhavya Tiwari (24csu040), Eklavya (24csu059), Bhavya Sangwan (24csu039), Pulkit Kush (24csu0166), Rishika Sharma (24csu0173)

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